ML algorithms and metrics

Which tool to pick, and how to tell if it works.

TODAY'S PLAN

Data is ready. Now we learn from it

Yesterday in one line: Good models start with good, honest, well-prepared data. Today: the learners that eat that data, and the yardsticks that judge them.

1

Three families of ML

Supervised, unsupervised, semi-supervised.

2

Meet the algorithms

Lines, neighbours, trees, clusters.

3

Judge a model

Overfitting, accuracy, precision, recall.

4

Pick the right tool

Which algorithm and metric, when.

THE BIG THREE

How much of an answer key do we have?

Supervised

Every example comes with its answer. The model learns to reproduce answers.

Spam or not? House price?

Unsupervised

No answers at all. The model finds structure on its own.

Customer groups? Odd one out?

Semi-supervised

A few answers, plus lots of unlabelled data to learn the shape.

100 tagged photos, 100,000 raw

A fourth family exists: **reinforcement learning** (learning by trial and reward, like a game-playing AI).

SUPERVISED LEARNING

Two kinds of answers: a category or a number

Classification

Predict a category.

- email → spam or not spam
- photo → cat, dog or fox

Regression

Predict a number.

- house → price ($240,000)
- weather → temperature (27°C)

WHICH IS IT?

Will it rain tomorrow (yes or no)? → Classification

Which language is this message written in? → Classification

How many millimetres of rain? → Regression

How many minutes until delivery? → Regression

ALGORITHM 1 · REGRESSION

Linear regression: the best straight line

Draw the line that sits closest to all the points. Then read a prediction off it.

```
price = w × size + b
w = slope, b = starting level
```

- **Learns via:** Mean squared error + gradient descent.
- **Good for:** Prices, sales, temperatures. Easy to explain.
- **Weak when:** The pattern is curved.



Size (m²) vs Price
New 100 m² house → About $185k

Logistic regression: probability & decision

Despite the name, it is a classifier. It outputs a probability between 0 and 1.

Hours studied vs Pass probability:

5 hours → 50% chance | 8 hours → 95% chance

Decision rule: Above the cut-off (e.g. 0.5) say PASS, below say FAIL. Cut-off can be adjusted.

Good for: Yes/no questions with probabilities (spam, churn, pass/fail).



S-Curve Sigmoid Output

Cut-off = 0.5

ALGORITHM 3 · CLASSIFICATION

k-Nearest neighbours: ask the neighbours

Step 1

Pick k, say **k = 3**.

Step 2

Find the 3 closest points using distance.

Step 3

Let them vote: 2 Teal, 1 Coral → **Group A**.

Good for

Small data, recommendations, a quick baseline.

Watch out

Scale your features. Slows down on huge dataset.

No "Training"

It simply remembers all the examples.

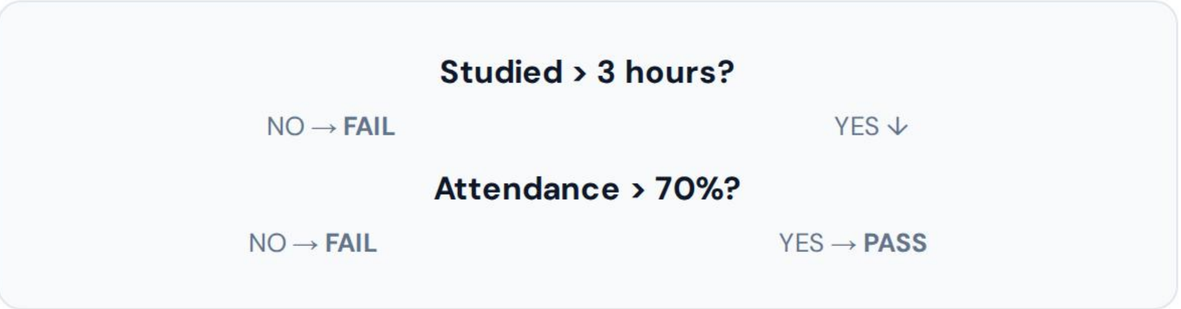
ALGORITHM 4 · EITHER KIND

Decision trees: a flowchart the machine writes

The model learns which questions to ask, in which order, to sort examples into the right answer.

Big Plus: You can read the rules. Banks and hospitals like that.

Big Minus: One deep tree memorises data. That is called *overfitting*.



MORE SUPERVISED TOOLS

Four more tools in the toolbox

Random forest

Hundreds of different trees vote. The crowd is wiser than one tree. A strong default for tabular data.

Naive Bayes

Bayes' idea: each word is a clue that shifts the odds. Fast and great for spam and text classification.

Support vector machine

Draws the widest possible "street" between two groups. Works well on small, clean datasets.

Neural networks

Layers of tuned weights. Need lots of data, but master images, sound, and language (ChatGPT).

UNSUPERVISED · CLUSTERING

k-Means: find the groups by yourself

1

Choose k, how many groups (e.g., k = 3).

2

Drop 3 centres at random in space.

3

Colour each dot by its nearest centre.

4

Move centre to middle of color. Repeat.

Use it for: Grouping shoppers, news articles, songs. No labels needed.

Catch: You choose k, and a human names the discovered groups.

UNSUPERVISED · MORE JOBS

No labels, many useful jobs

Clustering

Group similar things together. Customer types, news topics.

Anomaly detection

Spot the odd one out. Financial fraud, broken factory machinery.

Dimensionality reduction

Squash 100 columns into 2 so humans can plot and see them.

Association rules

"People who buy bread often buy butter." Shop shelf layouts.

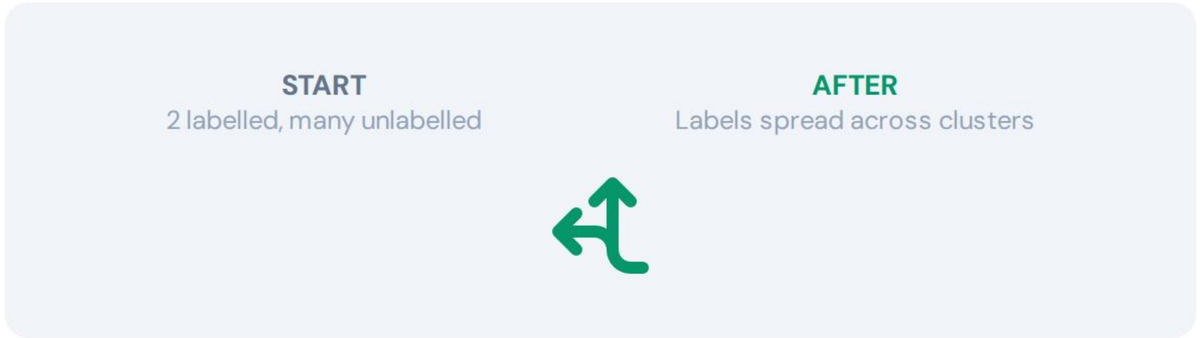
SEMI-SUPERVISED LEARNING

A few labels can label the rest

The Core Idea

Unlabelled dots show the underlying groups or structure. A few initial labels spread to name the remaining cluster.

> **Real use case:** 200 doctor-labelled medical scans combined with 20,000 unlabelled scans.



IS IT ANY GOOD?

Overfitting: memorising is not learning

Underfit

Too simple. Misses the underlying pattern entirely.

Just Right

Follows the general trend, ignores random noise.

Overfit

Memorised every single dot, noise included.

The Tell-Tale Sign: Great performance on training data, poor on new data.

The Cure: Hold back about 20% of data as an unseen test set.

METRICS

The confusion matrix: 4 ways to be right/wrong

A spam filter checks 100 emails (10 real spam, 90 normal):

TRUE POSITIVE (6)
Caught real spam

FALSE NEGATIVE (4)
Missed spam (In inbox)

FALSE POSITIVE (2)
False alarm (Normal flagged)

TRUE NEGATIVE (88)
Correctly left alone